RAIV-RML Joint Workshop

Environment Mapping

LiDAR SLAM + Monocular depth

JULY 09, 2026

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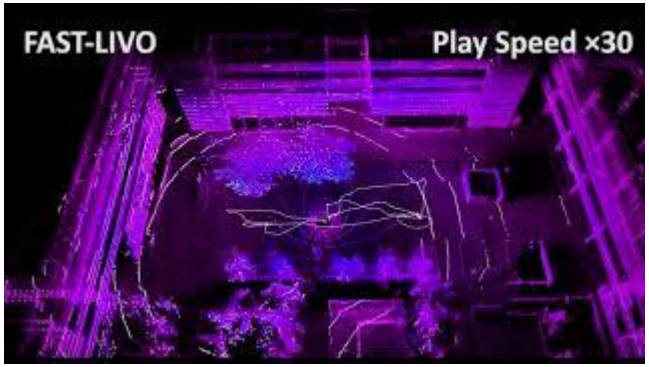
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1. Introduction

1.1. Purpose





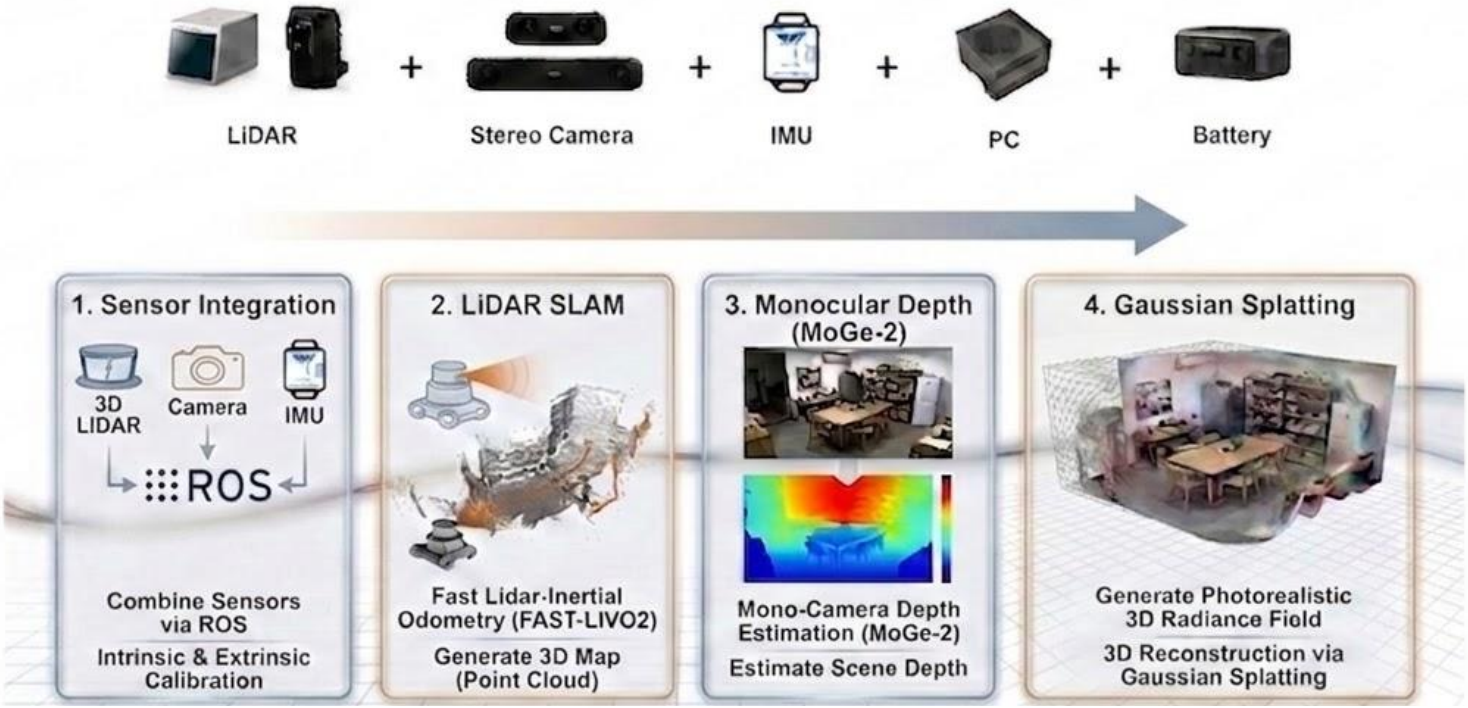
This project aims to build a robust 3D environment map for autonomous systems.

Goal: **Handheld Sensor Systems** for Data Acquisition.

Critical Issue: **LiDAR sensors fail on transparent objects** (laser penetration/scattering).

Our Approach: Compensating LiDAR limitations **using AI-based Camera Depth Estimation.**

1.2. System Pipeline



2. LiDAR SLAM

2.1. Related Works: LiDAR SLAM vs Visual SLAM

LiDAR SLAM

▲ Strengths

Provides precision-driven 3D mapping and high environmental robustness.

▼ Weaknesses

Remains physically blind to transparent objects like glass surfaces.

Visual SLAM

▲ Strengths

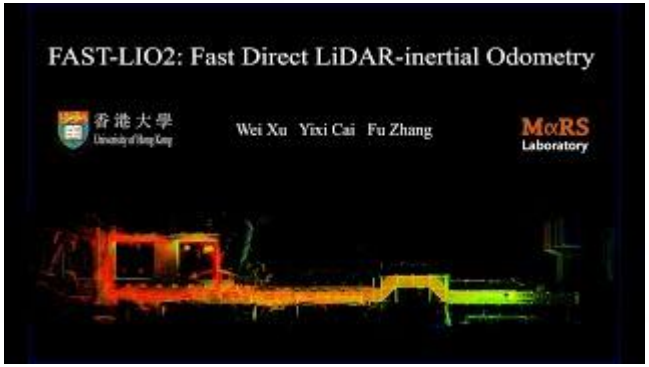
Cost-effective and lightweight framework utilizing rich visual pixel data.

▼ Weaknesses

Suffers from low depth estimation accuracy and high sensitivity to illumination changes.

We chose a LiDAR-centric framework to secure a highly reliable basemap, **strategically using vision** to compensate for LiDAR's blind spots.

2.1. Related Works: LiDAR SLAM



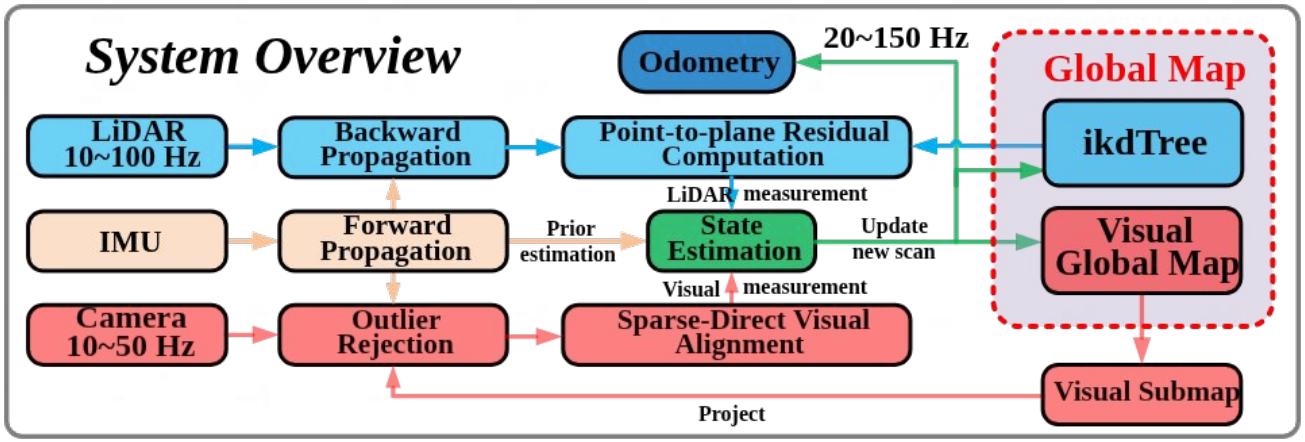


How it Works:

1. **Preprocessing:** Filtering and downsampling raw sensor data.
2. **Odometry Estimation:** Continuously tracking 3D position and rotation (6-DoF).
3. **Global Mapping:** Aligning registered scans into a unified 3D coordinate.

Key Performance Indicator: Maximizing Odometry Accuracy to eliminate accumulation errors (Drift) without relying on heavy correction algorithms.

2.2. FAST-LIVO2



FAST-LIO2 drastically reduces computational load while increasing state estimation speed.
FAST-LIVO2 paints the 3D point cloud map with real-world colors and **survives extreme environments** where LiDAR alone might fail.

📡 **LiDAR:** Builds the 3D geometric skeleton (Depth & Shape).
⚖️ **IMU:** Tracks rapid motions and maintains the system's orientation.
📷 **Camera (Vision):** Provides RGB textures and acts as a "backup eye" in feature-less areas.

2.3. Hardware Integration

Hardware Modules

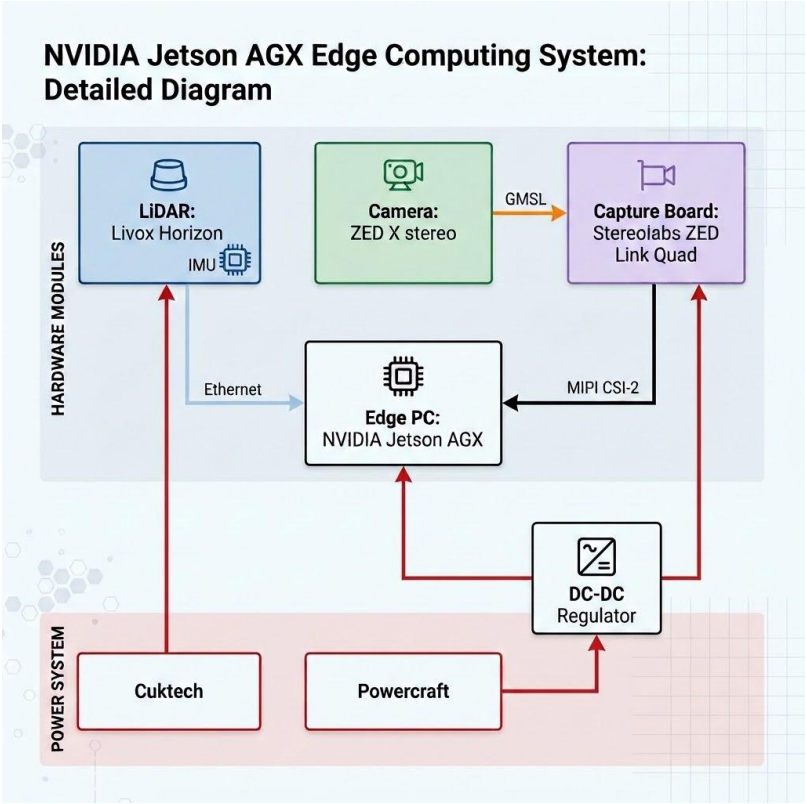
- Edge PC: NVIDIA Jetson AGX
- LiDAR & IMU: Livox Horizon
- Camera: ZED X stereo
- Capture Board: Stereolabs ZED Link Quad

Connection Interfaces

- LiDAR: Ethernet
- Camera: GMSL

Power System

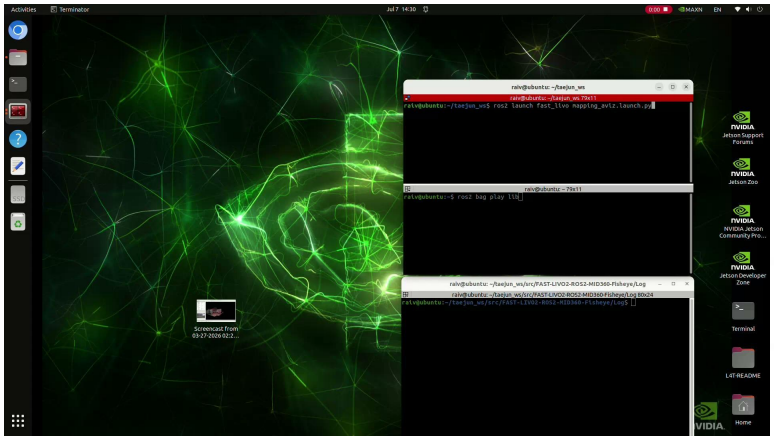
- LiDAR: Cuktech
- Edge PC & Capture Board: Powercraft
- Custom Circuit: DC-DC regulator for stable voltage during inference



2.4. Demonstration





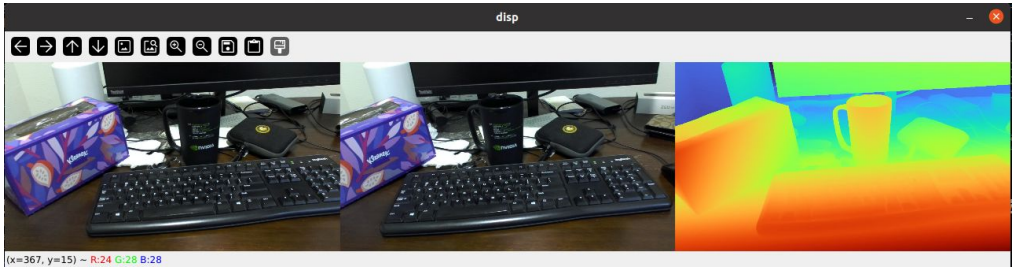


The physical disparity between the Camera and LiDAR axes is minimized, and pitch and roll corrections are applied for precise line-of-sight matching.

Mapping via FAST-LIVO: highly accurate on solid walls, but **missing transparent surfaces.**

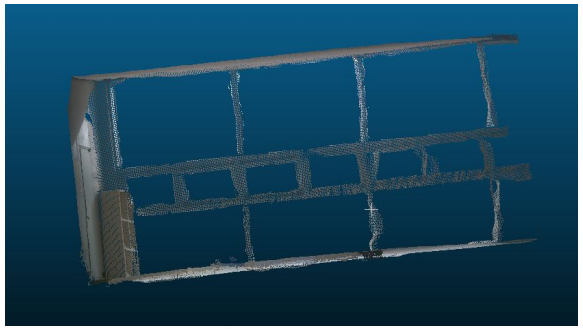
3. Monocular depth

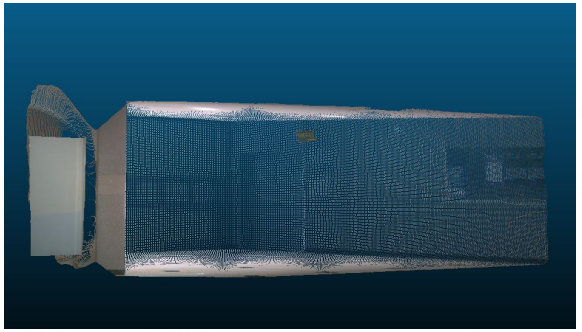
3.1. Related Works: Stereo depth Estimation





Stereo vision(FoundationStereo) calculates highly accurate geometric distances **using pixel disparity between two cameras**,

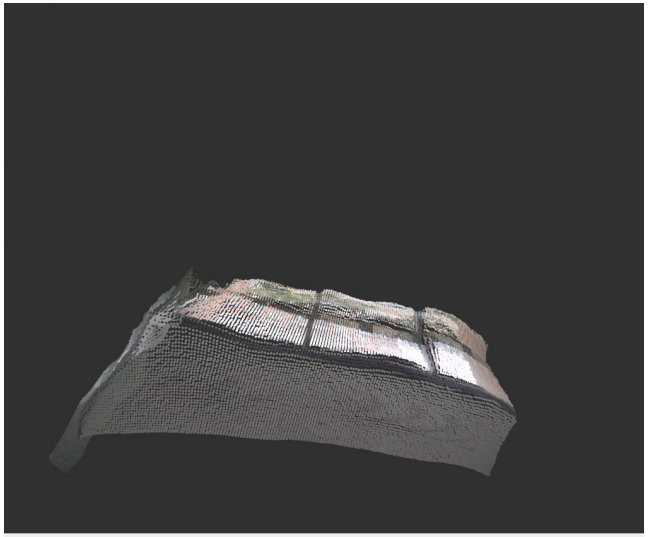




but fundamentally fails on texture-less surfaces like transparent glass.

3.1. Related Works: Monocular depth Estimation



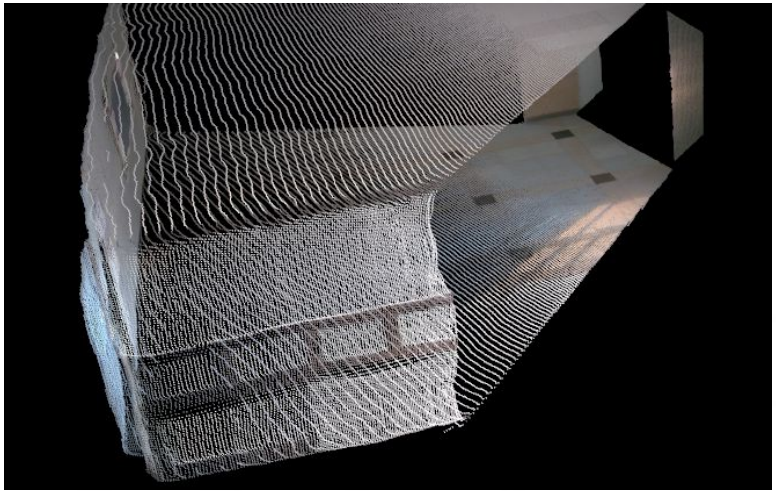


Monocular depth utilizes AI to **analyze the overall context of a single image**, **successfully inferring the depth and global scale of objects** even where geometric matching fails.

Depth-Anything-V3: Suffers from scale expansion and "tunnel" distortion.

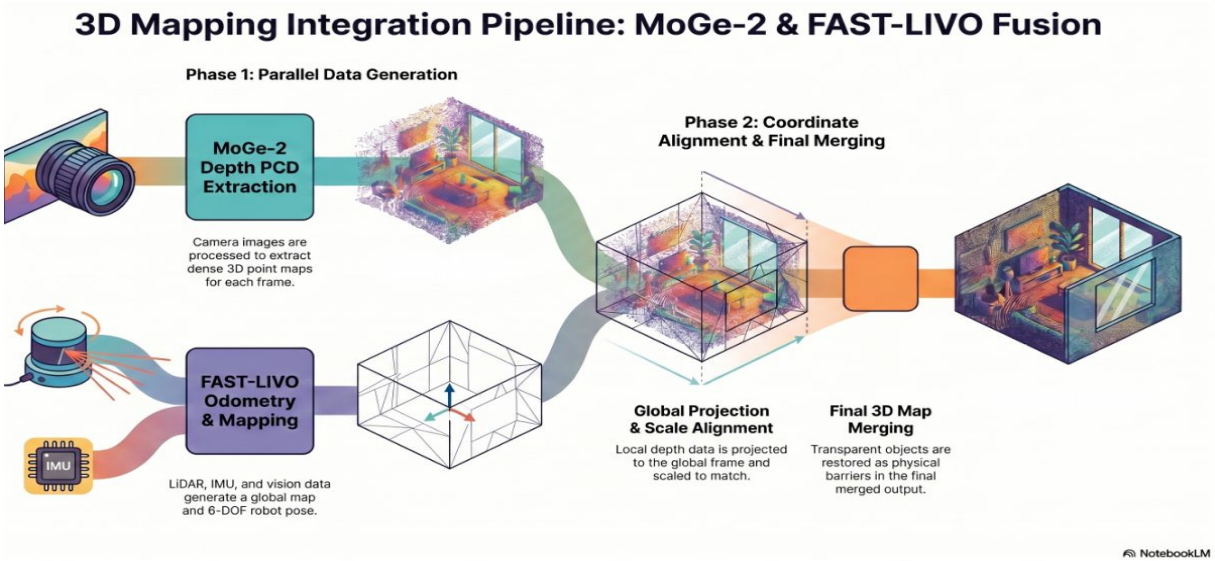
3.2. MoGe-2

MoGe-2



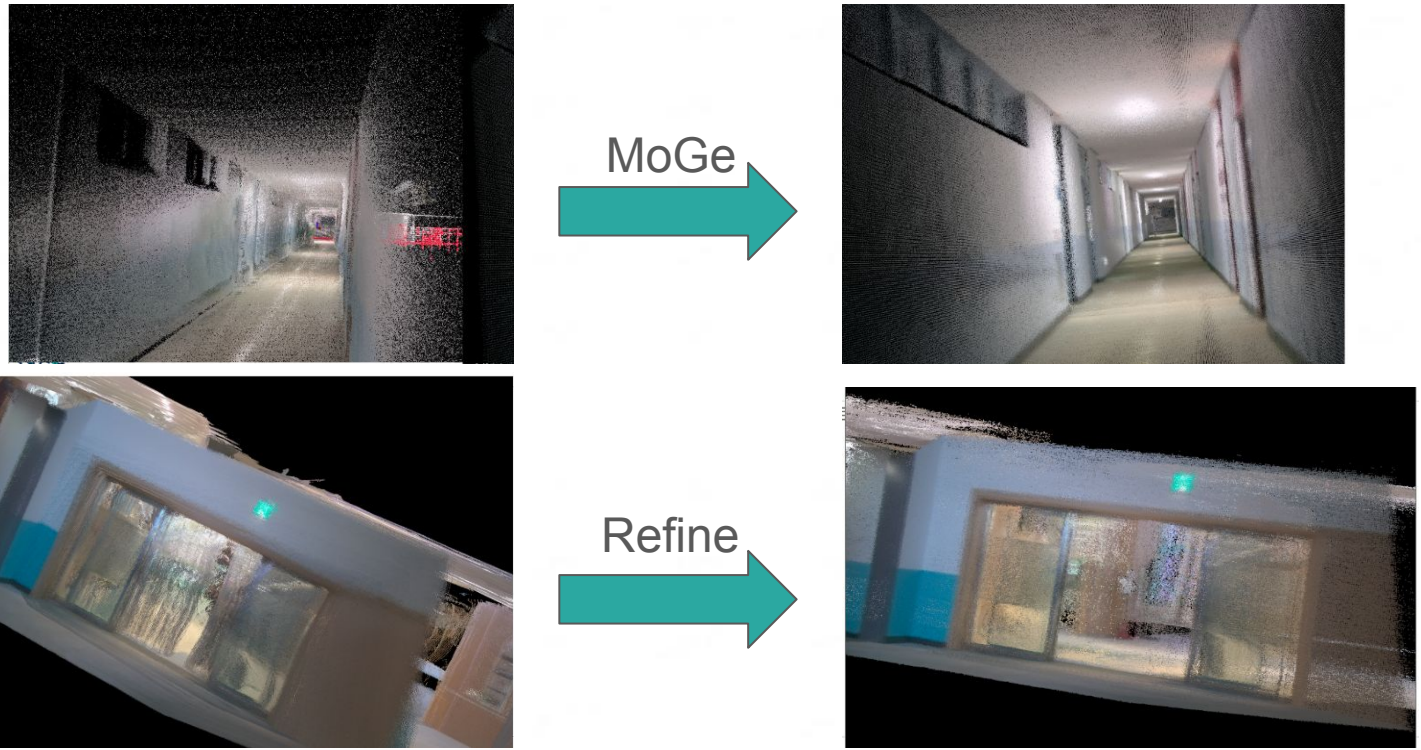
Successfully reconstructs 3D geometry without severe distortion.

3.3. Fusion: FAST-LIVO2 + MoGe-2

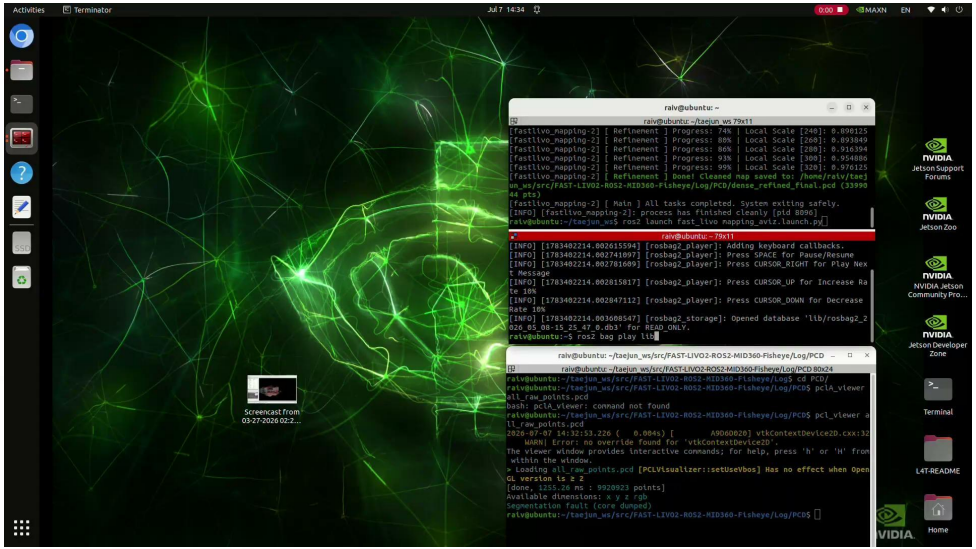


Phase 1: Extract 6-DoF Odometry (FAST-LIVO2) & Depth PCD (MoGe-2).
Phase 2: Global projection of local depth points using odometry.
Phase 3: Scale alignment and noise filtering based on LiDAR plane references.

3.3. Fusion: FAST-LIVO2 + MoGe-2



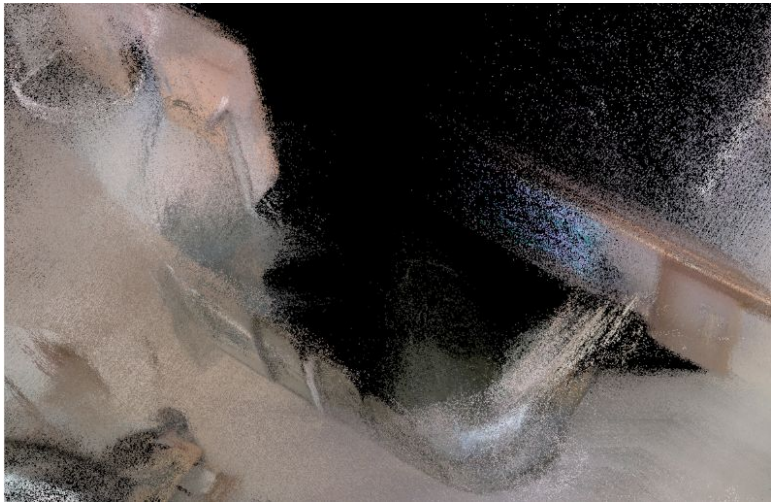
3.4. Demonstration

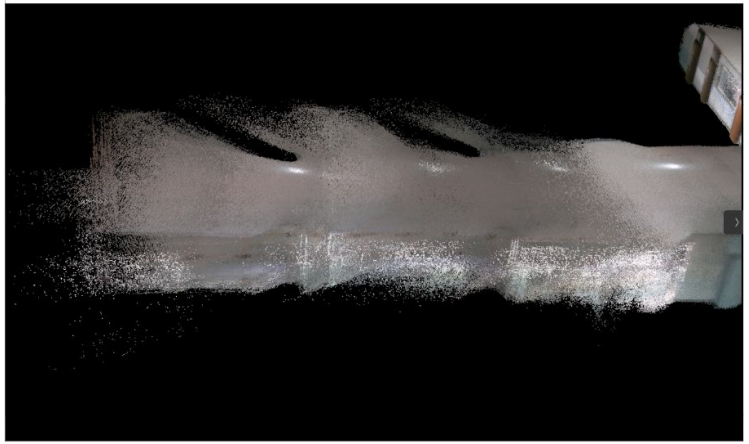


Mapping via Fusion: **Restoring missing transparent barriers.**

4. Future Work & Conclusion

4.1. Future Works: Remaining Glass Issues

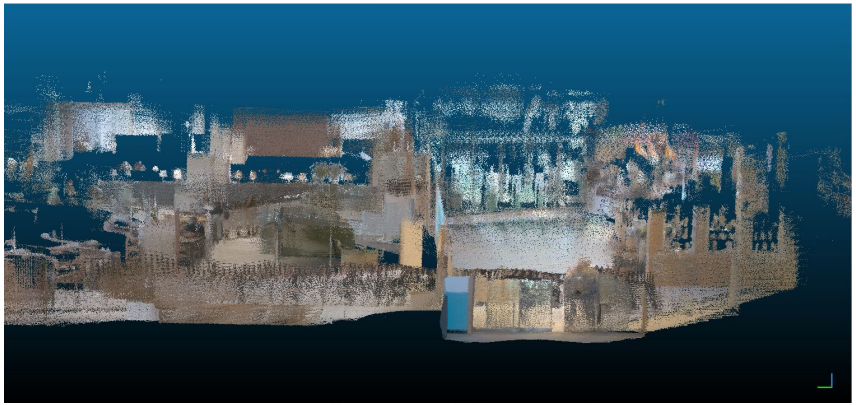


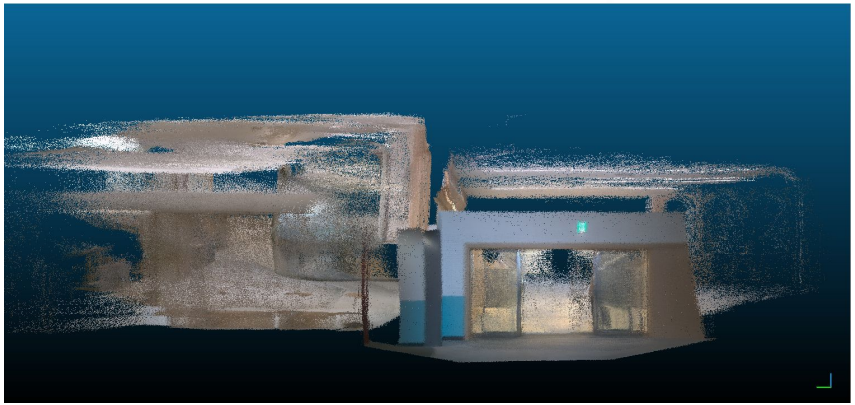


Remaining Challenges: Cylindrical glass curvature fitting & background depth projection.
Next Step: **Integrating 2D Semantic Segmentation (e.g., Grounded-SAM 2 / LLMDet)** to isolate precise glass masks.

Final Goal: Photorealistic 3D environment rendering. Applying 3D Gaussian Splatting on the fully refined point cloud map.

4.2. Conclusion





Successfully integrated AI Monocular Depth into the C++ LiDAR SLAM backend.
Overcame the physical limitations of LiDAR on transparent surfaces.

Expected to achieve a dense, robust 3D map ready for advanced autonomous navigation.